

SHAUNAK RANE

AI/ML Engineer | Backend & LLM Systems | Python | FastAPI

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TECHNICAL SKILLS

Languages & Core: Python, C, SQL, Bash, Linux, Git

AI & LLM Systems: Multi-Agent Workflows, RAG Architecture, Tool & Function Calling, Structured Outputs (JSON Schema), Vector Embeddings (pgvector), Prompt Engineering

Machine Learning & Data: PyTorch, PyTorch Geometric, GraphSAGE, Scikit-learn, Statsmodels, NumPy, Pandas, Time-Series Modeling, Anomaly Detection, Cross-Validation

Backend & Cloud/DevOps: FastAPI, Flask, RESTful APIs, WebSockets, PostgreSQL, Supabase, MySQL, Docker, pytest, CI/CD, Postman

WORK EXPERIENCE

Univitt Technologies - AI Engineering Intern

May 2026 - Aug 2026 | Thane, India (Remote)

- Engineered predictive Condition-Based Monitoring (CBM) pipeline processing 4,000+ daily historian records across 3 industrial facilities, reconciling multi-stage thermodynamics.
- Reproduced 3,879 historical Degradation Factor Index (DFI) records to $<1e-10$ absolute error against plant baseline; formulated causal maintenance-reset logic validated across 17 automated regression gates.
- Containerized Flask REST inference microservices with Docker, delivering sub-second anomaly detection and thermodynamic fouling score endpoints.
- Audited 400+ physical and mechanical process variables, evaluating ML models against Holt damped-trend statistical baselines; instituted automated CV promotion gates that halted ungrounded maintenance alerts.

Univitt Technologies - Software Engineering Intern

Apr 2025 - Jul 2025 | Thane, India (Remote)

- Developed RESTful backend APIs and relational database schemas with FastAPI and PostgreSQL for an institutional food operations and cost-management platform (serving 1,200+ daily meals).
- Automated weekly menu planning and multi-tier Bill-of-Materials (BOM) ingredient rollups, eliminating manual inventory calculation overhead.
- Trained a Random Forest meal-attendance forecaster incorporating weather, calendar events, and shift historical trends, achieving 88% accuracy (within $\pm 10\%$ headcount margin) and outperforming rolling-average baselines by 14%.
- Maintained 85%+ backend test coverage with pytest, executed Alembic database schema migrations, and collaborated on code reviews using Git pull-request workflows.

PROJECTS

Project Aegis - Autonomous Multi-Agent Threat Intelligence System Python, FastAPI, Multi-Agent Swarms, Agentic AI, RAG, Supabase (pgvector), Docker

Live Demo: aegis-protocol-110.pages.dev | **Source Code:** github.com/Sh Shaunakrane914/Misinformation

- Architected a decoupled two-stage agentic swarm (Research Agent $\rightarrow$ Investigator Agent) enforcing strict JSON Schema validation, achieving 100% structured parsing reliability across model retries.
- Engineered quantitative market anomaly detection (Scout Agent) computing 5-day rolling Z-scores on price/volume spikes via Yahoo Finance proxies, distinguishing organic volatility from coordinated disinformation attacks.
- Implemented canonical SHA-256 claim deduplication to eliminate redundant LLM token spend (0ms cached recall), coupled with Supabase pgvector semantic retrieval and auditable verdict logs.

TopoFlow - GNN Permeability & Micro-CT Pore Network Benchmark

Python, PyTorch Geometric, GraphSAGE, OpenPNM, FastAPI

Source Code: github.com/Sh Shaunakrane914/Flow

- Benchmarked GraphSAGE against classical Kozeny-Carman physics baselines across 1,231 micro-CT pore networks from 5 geological formations, establishing an empirical regime threshold (Cv) for when graph topology outperforms bulk equations.
- Achieved 46.2% MSE reduction on heterogeneous Savonnières carbonate and 28.4% on Estailledes formations over classical solvers, while identifying that homogeneous sandstones remain better predicted by classical physics.
- Built end-to-end pore-network extraction pipeline (PoreSpy/SNOW2), GNN model training workflows in PyTorch Geometric, and streaming FastAPI inference endpoints with SSE.

Gridium Protocol - Autonomous Microgrid Optimization & DePIN Market

Python, PyTorch, DDPG, FastAPI, WebSockets, Solidity, Circom

Live Demo: live-ai-1-7tcy.vercel.app | **Source Code:** github.com/Sh Shaunakrane914/Live-Ai-1

- Developed a continuous-control reinforcement learning engine (PyTorch DDPG) in a custom 15-node Ohm's law Gymnasium microgrid (AegisEnv), dynamically adjusting AMM swap fees (0.10%–5.00%) to mitigate solar duck-curve volatility.
- Reduced peak-hour grid strain by 18.4% compared to static flat-fee and rule-based fixed-tariff baselines across simulated multi-node solar generation and battery demand cycles.
- Engineered 500ms Socket.io telemetry gateway, Solidity constant-product AMM contract ($x \times y = k$) with reentrancy protection, and Groth16 zk-SNARK circuits proving energy surplus off-chain.

EDUCATION

Universal AI University - B.Tech in Artificial Intelligence & Machine Learning

Expected May 2028 | Karjat, Maharashtra

- CGPA: 8.47 | Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Deep Learning, DBMS, Operating Systems, Linear Algebra & Probability.